

FMCG Multi-Country Sales Dataset Analysis

(Project Brief)

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1.1 Project Overview

This project is an end-to-end FMCG Sales Analytics project developed using a synthetic three-year FMCG sales dataset containing approximately 1.1 million records. The dataset captures daily sales and operational information across stores, products, categories, brands, geographic locations, sales channels, suppliers, promotions, pricing, inventory, and selected external factors such as weather and holidays.

The project aims to transform the raw sales data into a structured analytical solution through a complete data analytics workflow. The process begins with data profiling, quality assessment, cleaning, and validation, followed by relational database development in MySQL, SQL-based business analysis, and interactive visualization in Power BI.

The analysis focuses on understanding sales performance, product and category contribution, geographic and store-level performance, channel performance, promotional effectiveness, profitability, inventory availability, stockout exposure, and demand characteristics. The final outcome will translate these findings into actionable business insights and recommendations that could support commercial, pricing, promotional, and inventory-related decision-making in an FMCG environment.

1.2 Business Problems

FMCG businesses generate large volumes of transactional and operational data across products, stores, geographic markets, sales channels, suppliers, pricing, promotions, and inventory. When this information is stored as a large raw dataset, it can be difficult for business stakeholders to identify the major drivers of revenue, sales volume, profitability, and product availability.

Management needs to understand which products, categories, brands, stores, markets, and channels are performing well and which areas require improvement. In addition, promotional activities and discounts need to be evaluated to determine whether increased sales volume is accompanied by sufficient commercial value. Inventory-related issues such as stockouts can also affect the ability to meet customer demand and may result in lost sales opportunities.

Therefore, the business problem addressed by this project is to convert a large and initially unstructured FMCG sales dataset into a reliable analytical model that can provide clear answers to key commercial and operational questions, identify performance gaps and opportunities, and support data-driven business decision-making.

1.3 Project Objectives

- i. Assess the quality, consistency, completeness, and structural integrity of the raw dataset.
- ii. Clean and prepare the dataset for analytical use.
- iii. Understand the dataset's business grain, dimensions, measures, and underlying relationships.
- iv. Develop a structured relational analytical database using MySQL.
- v. Perform descriptive, comparative, diagnostic, and trend-based analysis using SQL.

- vi. Analyze revenue, sales volume, pricing, discounts, promotions, margins, inventory, stockouts, and demand characteristics.
- vii. Identify high-performing and underperforming products, categories, brands, stores, markets, and sales channels.
- viii. Evaluate promotional and discount patterns in relation to sales performance.
- ix. Identify potential profitability and inventory-management opportunities.
- x. Develop an interactive Power BI dashboard to communicate key findings effectively.

1.4 Dataset Source

Dataset Website: Kaggle.com

Title: FMCG Multi-Country Sales Dataset

File Name: *fmcg_sales_3years_1M_rows.csv*

File Size: 202 MB

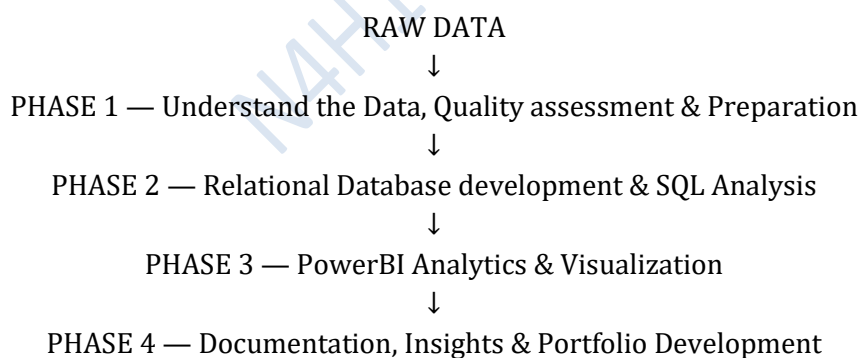
Dataset link: <https://www.kaggle.com/datasets/robertocarlost/fmcg-multi-country-sales-dataset>

1.5 Dataset Components

The dataset contains one CSV file: “**fmcg_sales_3years_1M_rows.csv**” - a 3-year sales record (from 1-Jan-2021 to 31-Dec-2023) of 15 countries and 60+ cities, having 1.1M+ business records. With over 350+ SKU, seasonality patterns, promotions & segmentation data – this dataset is designed for real-life business cases, job interviews, portfolio projects, coursework, and professional S&OP simulations.

1.6 Project Progression

These are the phases followed to complete this project –



1.7 Tools & Technologies

Operational tools used for this project: MS Excel, MySQL, PowerBI (web version), Notepad, Github.

However, as I’m going to use PowerBI online and the dataset lays in my local MySQL server, so to connect the database with PowerBI online version and maintain out-bound open connection, I need to set environment by installing – 1. On-premises Gateway (from Microsoft) & 2. MySQL Connector/NET 26.7.0

1.8 Data Limitations

Synthetic Dataset:

The dataset is simulated rather than collected from an actual FMCG company’s operational system.

Consequently, the distributions, relationships, and patterns observed in the data may not fully represent real-world FMCG business conditions.

Limited Business Context:

The dataset does not provide detailed information about actual company strategy, sales targets, competitor activity, commercial agreements, distribution policies, marketing expenditure, or management decisions. These factors may influence real-world sales performance.

Limited Customer-Level Information:

The available data primarily focuses on sales, products, stores, geography, channels, suppliers, promotions, and inventory. Detailed customer-level attributes and purchasing behavior are therefore outside the scope of this analysis.

Currency Considerations:

The dataset contains geographic and monetary information that may involve different currencies. Revenue and other monetary values across different currencies should therefore not be directly aggregated or compared without appropriate currency normalization.

Missing External Market Factors:

Real-world factors such as competitor prices, competitor promotions, advertising expenditure, market share, consumer demographics, economic conditions, and changes in distribution strategy are not comprehensively represented in the dataset.

Inventory Data Limitation:

Stock-on-hand and stockout indicators provide useful inventory-related signals but may not represent the complete inventory lifecycle, including warehouse-level inventory, replenishment orders, damaged products, returns, transfers, or lost sales.

1.9 Project Timeline

I targeted this project to be completed in 7 days. This was my objectives/schedules throughout the time period:

Days	Objectives	Outputs
Day 1-2	Understand, inspect, clean & prepare data	Clean datasets + data-quality report
Day 3-5	MySQL analysis + business insights + Power BI model	SQL analysis + analytical dataset + initial dashboard
Day 6-7	Complete Power BI + documentation + screenshots + GitHub	Complete Project Portfolio

Final documentation and portfolio organize in GitHub might take 2 more days.